

A Non-Monotonic $a_0(z)$ Signature: An Observational Discriminant Among Evolving-Acceleration-Scale Models in the DESI Era

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Abstract

The MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ is numerically close to $c\sqrt{\Lambda}$, a coincidence that has long invited a cosmological origin and, with it, the possibility that a_0 *evolves* with redshift. Several distinct couplings have been proposed, and — crucially — they disagree in the **shape** of $a_0(z)$, not merely its amplitude. This note catalogs the published $a_0(z)$ predictions and observes that they are, without exception, **monotonic**: rising ($a_0 \propto cH(z)$ in emergent-gravity and quantized-inertia models; holographic dark energy; the Λ CDM-simulation-inferred $a_0 \propto (1+z)^{3/4}$), constant (standard MOND; $a_0 \propto \sqrt{\Lambda}$ with Λ fixed), or monotonically declining (the Limbach–Psaltis–Özel 2008 relation $a_0 \propto \sqrt{\rho_{\text{DE}}}$ for a constant equation of state $w < -1$). Evaluating the LPO relation instead on the **DESI DR2 dynamical (w_0 – w_a) dark energy** — where $\rho_{\text{DE}}(z)$ is itself non-monotonic, peaking at the phantom-crossing near $z \approx 0.4$ — yields a qualitatively different curve: a **bump** of +3–9% in a_0 at $z \approx 0.4$, followed by a decline to ≈ 0.65 – 0.78 by $z = 3$. This non-monotonic shape is absent from every rival prediction we catalog, and a thorough (though not exhaustive) literature search found no prior publication of it. It therefore constitutes a clean, falsifiable **observational discriminant**: at low z it mimics a rising model, at high z a declining one, and only it exhibits the turnover. We are explicit about credit and scope. The relation $a_0 \propto \sqrt{\rho_{\text{DE}}}$ is Limbach, Psaltis & Özel (2008); the de Sitter–Unruh modified-inertia mechanism is Milgrom (1999), with a concurrent 2026 treatment by Luo; the ρ_{DE} peak is standard DESI w_0 – w_a cosmology. The contribution here is the **composition** — the DESI-era evaluation and the identification of the resulting non-monotonic signature as a discriminant — together with an honest assessment that the bump lies below current detection floors and is therefore a long-horizon test.

1. One relation, several couplings, several shapes

That a_0 is comparable to cH_0 and to $c\sqrt{(\Lambda/3)}$ is an old observation (Milgrom 2001; Famaey & McGaugh 2012). If it reflects a genuine coupling, a_0 should track its cosmological source through cosmic time, and the *form* of the source fixes the *shape* of $a_0(z)$. Two candidate couplings dominate the literature:

$$a_0(z) \propto cH(z), \quad a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}.$$

The first ties a_0 to the expansion rate; the second to the dark-energy density. Limbach, Psaltis & Özel (2008) wrote both explicitly (their Eqs. 3–4) and proposed testing them with the redshift evolution of the Tully–Fisher intercept, using $M = V^4/(G\alpha_0)$. Because $H(z)$ and $\rho_{\text{DE}}(z)$ have different redshift dependences, the two couplings predict different $a_0(z)$ — and this is exactly the leverage a high- z Tully–Fisher or radial-acceleration measurement provides.

The point of the present note is narrower and, we believe, not previously made: **once dark energy is allowed to be dynamical, the $\sqrt{\rho_{\text{DE}}}$ coupling predicts a shape that no other model produces.**

2. The landscape of published $a_0(z)$ predictions — all monotonic

We surveyed the published predictions and measurements of a redshift-dependent MOND scale. They sort cleanly by shape (Table 1), and every one is monotonic or flat.

Rising (a_0 larger in the past): emergent gravity and quantized inertia with $a_0 \propto cH(z)$ (Verlinde 2016; McCulloch 2017), which rise monotonically because $H(z)$ does; holographic dark-energy constructions in which ρ_{DE} tracks $H^2(z)$ (Adolf et al. 2024); and the Λ CDM-hydrodynamical-simulation result that the *fitted* a_0 rises as $\approx(1+z)^{3/4}$ from baryon-fraction evolution (Mayer et al. 2023). The intermediate- z kinematic measurements (MUSE-DARK III, Ciocan et al. 2026; MIGHTEE-HI, Vărășteanu et al. 2025) also report a *linearly rising* fitted a_0 — a monotonic datum, not a peaked prediction.

Constant: standard Milgrom MOND, and constructions in which $a_0 \propto \sqrt{\Lambda}$ with Λ a true cosmological constant (Milgrom 2015; Blanchet & Serraille 2025). A recent de Sitter–Unruh modified-inertia treatment (Luo 2026) shares the mechanism used here but yields a *static* a_0 with no dark-energy-density evolution.

Monotonically declining: the LPO 2008 $\sqrt{\rho_{\text{DE}}}$ coupling for a constant equation of state $w < -1$, for which $\rho_{\text{DE}} \propto a^{-3(1+w)}$ falls into the past.

Non-monotonic (bump): the subject of this note; no other occupant found.

The one genuinely non-monotonic *rival* — the Scale-Invariant Vacuum theory (Maeder & Gueorguiev; Gueorguiev 2024) — is not a match: its turnover is driven by the matter density Ω_m through Weyl scale invariance, not by a dark-energy phantom crossing; it is not tied to DESI, w_0 – w_a , or $\sqrt{\rho_{\text{DE}}}$; and for standard $\Omega_m \approx 0.3$ its curve is in practice a monotonic decline with no $z \approx 0.4$ peak.

3. The non-monotonic signature

Take the LPO relation $a_0(z) = a_0(0) \cdot [\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0)]^{1/2}$ and feed it the DESI DR2 dynamical dark energy in the CPL form $w(a) = w_0 + w_a(1-a)$, for which

$$\frac{\rho_{\text{DE}}(z)}{\rho_{\text{DE}}(0)} = (1+z)^{3(1+w_0+w_a)} \exp\left(-\frac{3w_a}{1+z}\right).$$

For the DESI-preferred region ($w_0 > -1$, $w_a < 0$), $w(z)$ crosses -1 near $z \approx 0.4$, so $\rho_{\text{DE}}(z)$ **peaks there** — this is standard DESI cosmology. Propagating that peak through the square root gives $a_0(z)$ that **rises to a bump and then falls**:

DESI DR2 fit	bump in a_0	z of peak	$a_0(z=3)/a_0(0)$
DESI+CMB+DESY5	+6.0%	0.35	0.65
DESI+CMB+Union3	+8.9%	0.44	0.71
DESI+CMB+Pantheon+	+3.6%	0.36	0.78

At low redshift the curve is indistinguishable from a rising model; at high redshift from a declining one; **only the $\sqrt{\rho_{\text{DE}}}$ -on-dynamical-DE curve turns over**. The turnover location is not free — it is pinned to the phantom-crossing redshift, itself measured by DESI.

4. Why the bump is structurally unique

Every rival maps a_0 onto a *monotonic* driver: $H(z)$, a power of $(1+z)$, or a constant- w power-law ρ_{DE} . A bump requires $\rho_{\text{DE}}(z)$ *itself* to be non-monotonic, and that requires an equation of state that crosses $w = -1$ with the DESI sign. The two ingredients each have prior art — the $\sqrt{\rho_{\text{DE}}}$ relation (LPO 2008) and the ρ_{DE} peak (DESI w_0 - w_a cosmology) — but their **composition into an $a_0(z)$ prediction for galaxy dynamics** appears not to have been made. That composition is the entire content of the signature, and it is enough to make it a discriminant: no monotonic-driver model can reproduce a turnover.

5. Detectability — an honest floor

The distinctiveness is qualitative; the amplitude is not favorable. The bump translates to a Tully–Fisher/BTFR intercept shift of only ≈ 0.015 – 0.037 dex for the robust DESI fits ($a_0 \propto 1/M$ at fixed V), against systematic floors of ≈ 0.05 dex (star-formation) to ≈ 0.20 dex (spatial resolution; cf. the seeing systematic reported in intermediate- z JWST rotation-curve work, 2026). In weak-lensing the bump is ≈ 2 – 3% versus a ≈ 3 – 5% circumgalactic-gas floor; in direct kinematic $a_0(z)$ it is far below the ≈ 0.13 – 0.17 dex per-point scatter of current measurements. The signature is therefore a **long-horizon** discriminant — realistically requiring sub-2% Tully–Fisher-intercept precision at $0.3 < z < 0.6$ with star-formation and seeing systematics controlled — not a near-term test. The larger high- z *decline* is more accessible in principle but is the LPO direction, not distinctive, and demands clean deep-MOND kinematics at $z \gtrsim 2$ that do not yet exist.

6. Scope and credit

This note claims neither a new relation nor a new mechanism. The coupling $a_0 \propto \sqrt{\rho_{\text{DE}}}$ and its Tully–Fisher test are due to **Limbach, Psaltis & Özel (2008)**; the de Sitter–Unruh modified-inertia reading of MOND is due to **Milgrom (1999, 1994)**, with a concurrent 2026 treatment by **Luo**; the peak of $\rho_{\text{DE}}(z)$ is standard **DESI** dynamical-dark-energy phenomenology; and the high- z decline is already present in LPO for $w < -1$. The acceleration-scale coefficient used for normalization, $a_0 = c^2\sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m/s}^2$, is a one-parameter modelling choice, not a derivation. The contribution is the **DESI-era composition** and the recognition of the resulting **non-monotonic curve as an observational discriminant** among the $a_0(z)$ hypotheses. Our search for prior publication of this signature was thorough but not exhaustive: forward literature search across the DESI×MOND intersection and the full $a_0(z)$ landscape returned no match, but

a complete citation-tree enumeration of LPO 2008 could not be machine-verified, so we state the result as “no duplicate found,” not “none exists.”

References

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The author retracted all earlier “theory of everything” claims (2026-06-23); this note makes none. It is a comparative-predictions note that credits the originating relation (LPO 2008) and mechanism (Milgrom 1999) and claims only the DESI-era non-monotonic signature as a discriminant.